

The examination is being carried out on the **following application documents**

Description, Pages

1-305 as published

Claims, Numbers

1-15 filed in electronic form on 05-02-2021

1 Reference is made to the following documents:

- D1 WO 2017/140778 A1
- D2 SARAH DOYLE ET AL: "Targeting the NLRP3 inflammasome in chronic inflammatory diseases: current perspectives", JOURNAL OF INFLAMMATION RESEARCH, 1 January 2015 (2015-01-01), page 15, XP055555633, DOI: 10.2147/JIR.S51250
- D3 REBECCA C COLL ET AL: "A small-molecule inhibitor of the NLRP3 inflammasome for the treatment of inflammatory diseases", NATURE MEDICINE, 16 February 2015 (2015-02-16), XP055303035, New York ISSN: 1078-8956, DOI: 10.1038/nm.3806

2 Independent claims

- compounds of formula (I)(claim 1)
- their pharmaceutical compositions (claim 10)
- their pharmaceutical uses (claims 11-12)

3 Amendments (Article 123(2) EPC)

The present set of claims is based on the original application as follows:

- claim 1 corresponds to original claim 1
- new claim 2 is directed to compounds as claimed in claim 1 wherein R1 is defined as disclosed in paragraphs [0104], [169], [173] and original claims 3-5, 7-8, 11-13, 14
- claim 3 is based on original claims 18-20

-claim 4 is based on original claims 23 paragraphs [312], [320], [270], [286], [229], [238], [249], [353], [412], original claims 46

-claim 5 is based on original paragraphs [455], [459], [464], [468], [472], [475], [483], [487], [490], [496], [500] and original claim 66

-claim 6 is based on original claims 67-68

-claim 7 is based on original claims 69 to 87

-claim 8 is based on claim 86

-claim 9 corresponds to claim 88

-claim 10 corresponds to original claim 102

-claim 11 corresponds to original claim 105 reworded such to comply with the requirements of Article 53(c) EPC

-claim 12 corresponds to original claim 108 reworded such to comply with the requirements of Article 53(c) EPC

-claim 13 corresponds to original claim 116 reworded such to comply with the requirements of Article 53(c) EPC

-claim 14 corresponds to original claim 117 reworded such to comply with the requirements of Article 53(c) EPC

-claim 15 corresponds to original claim 118 reworded such to comply with the requirements of Article 53(c) EPC

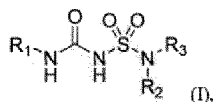
The amendments from the current set of claims are considered to fulfill the requirements of Article 123(2) EPC.

4 **Novelty** (Article 54 EPC)

The present compounds of formula (I) (independent claim 1) are a novel selection from the general structures disclosed in claims 1 and 7 (formula V) of D1 and differ from the compound MCC950 disclosed in D3 on account of the present R2R3-N- substituent. D2 does not disclose any specific compounds containing a sulphonyl urea group. The present claimed subject-matter directed to compounds of formula (I), their pharmacological compositions and uses appears to fulfill the requirements of Article 54 EPC.

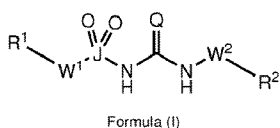
5 **Inventive step** (Article 56 EPC)

- 5.1 The present claimed compounds of formula (I) are useful to treat diseases connected to the inhibition of the NLRP3 activation.

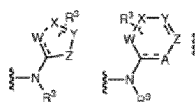


wherein R1 is a cycloalkyl or aryl, R2 is $-(CX_1X_2)_n-R_2s$ with $n=0-2$ and R_s =heterocycloalkyl and R3 is a heterocycloalkyl or heterocycloaryl

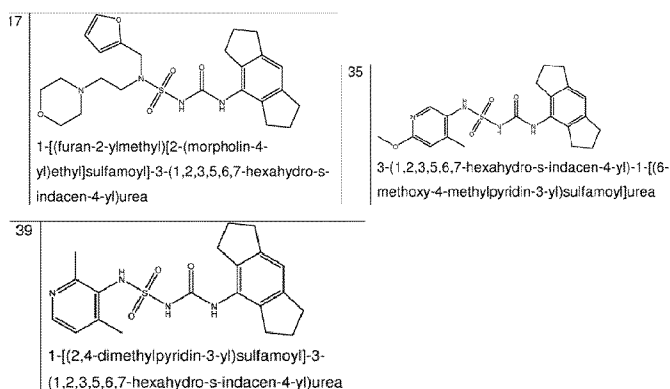
- 5.2 D1, which is regarded as being the closest prior art, discloses compounds of formula (I) with the same uses as the present ones.



The subject-matter of claim 1 can be considered as a selection of a formula (I) from the compounds of formula 1 or V disclosed in document (claims 1 and 7) for the same use. The skilled person starting from the compounds of formula (I) of claim 1 has to select the W2 to be absent, to limit the R2 to be cycloalkyl or aryl moiety, and R1 to be one heterocycloalkyl and the other one to be heterocycloalkyl or heterocycloaryl. Further, claim 7 (dependent on any of the previous claims) indicates as preferred structure the compounds of formula V, wherein R1W1 could be selected according to claim 5 to be (R1)2N, wherein at least one of R1 comprises a bicyclic group. Further, according to claim 6 the



R1W1 could be a group wherein R3 could be a heterocyclo or heteroaryl. As regards the specific examples disclosed in D1, compounds 17, 35 and 39 appear to be the closest structural for the present ones.



Thus, the present compounds of claim 1 represent a novel selection from the general structures disclosed in claims 5 to 7 of D1. The skilled person starting from examples 35, 39 and having in mind the technical problem of providing further sulfamoylureea derivatives useful to treat diseases connected to the inhibition of NLRP3 would expect that introducing of a substituent as defined for the present R2 (in particular in view of example 17 of D1) would not interfere with the claimed activity.

Thus, the above-mentioned technical of providing further sulfamoylureea derivatives useful to treat diseases connected to the inhibition of NLRP3 is considered to be solved in an obvious way and therefore it has to be reformulated as the provision of further sulfamoylureea derivatives useful to treat diseases connected to the inhibition of NLRP3 and having unexpected properties over the closest structural compounds disclosed in D1, namely the compounds 17, 35 and/or 39.

Since no such effects or properties are indicated and supported in the application, the subject-matter of claim 1 does not involve an inventive step within the meaning of Article 56 EPC. In order to show any such unexpected effect(s) vis-à-vis the said prior art, any comparative tests must be performed using compounds with the maximum structural similarity. This is necessary to show convincingly that the unexpected effect indeed originates from the distinguishing feature, in the present case the specific definition of R2 in combination with the definition of R3. In addition, it should be credibly shown that the effect occurs over the whole claimed range.

6 **Article 84 EPC**

- 6.1 Current claim 2 makes reference to any of the proceedings claims. This cannot be correct since the single preceding claim is claim 1.
- 6.2 Claim 9 does not fulfill the requirements of Article 84 EPC since it makes reference to the 166 compounds disclosed in the specification. According to Article 84 EPC, the claims have to be clear and concise, which is not the case for claim 9.

- 7 The Applicant is asked to file amendments which take account of the above comments and, where necessary, to harmonise the claims and the corresponding parts of the description. Article 123(2) EPC must be respected. The Applicant is asked to submit any corrections in the form of handwritten corrections to the application **as originally filed** and also in the form of retyped

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pages. For each amendment the Applicant is asked to indicate the support in the original disclosure. Terms as "incorporated by reference", "prodrugs" are to be deleted from the description and the methods of treatment have to be reworded to comply with the requirements of Article 53(c) EPC.